

Soil water redistribution processes around a tree on a hillslope: the effect of stemflow on the drying process

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ABSTRACT

Although many studies have investigated the ecological or hydrological influences of stemflow on soil water dynamics in the wetting process, the process by which stemflow affects soil water redistribution in the drying process has rarely been discussed. To address how soil water depletion occurs around a tree and the effects of stemflow on soil water depletion, we used a 2-year data set of soil water content to compare soil water depletion in the drying process before (SF period) and after (no-SF period) the interception of stemflow around a tree on a hillslope. The results revealed a dramatic soil water depletion at locations in the downslope area near to the tree, which was attributed to stemflow and root-induced pathways. The effect of stemflow on soil water depletion was significant in the short period (0.5 h) after the end of rainfall events, which increased the initial soil water content at the start of drying. The effect of root-induced pathways enhanced drainage during the drying process and resulted in irregular vertical distributions of changes in soil water content. Although soil water depletion at locations near to a tree would be expected to be greater than in locations farther away, due to root uptake in the drying process, most soil water depletion in the vicinity of a tree occurred in the SF period rather than the no-SF period. This was mainly attributed to greater instantaneous soil water depletion due to rainwater drainage rather than evapotranspiration, which occurred in the short period between rainfall events. This study furthers our understanding of the separate involvement of stemflow and root channels in the drying process in a warm temperate climate. We also suggest that the double-funnelling effect of a tree is valid not only for the wetting process but also for the drying process and can result in dramatic variations in soil water dynamics within the stemflow-influenced and root-influenced areas of a forested stand. Copyright © 2014 John Wiley & Sons, Ltd.

KEY WORDS double-funnelling effect; root channel; soil water content; soil water depletion; stemflow

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INTRODUCTION

Soil water distribution in a forested stand is uneven because of the effects of canopy interception and rainfall redistribution (Keim *et al.*, 2005; Keim *et al.*, 2006), root uptake (Schume *et al.*, 2004; Krämer and Hölscher, 2010) and other interactions (Tromp-van Meerveld and McDonnell, 2006; Archer *et al.*, 2012). Rainfall is first intercepted by canopies and then partitioned into throughfall and stemflow components as diffuse and point inputs, respectively. The soil moisture pattern may not necessarily match the diffuse pattern of throughfall (Pressland, 1976; Raat *et al.*, 2002), which could be affected by localized concentrations of stemflow input (Levia and Frost, 2003). Aboal *et al.* (1999) found that

precipitation could be concentrated up to 12.8-fold in the infiltration areas of the trees by stemflow in a laurel forest, even though the annual stemflow represented only 6.85% of the gross precipitation. Durocher (1990) recorded very rapid water movement and the generation of saturated zones underneath each of two sweet chestnut trees and attributed these features to small-scale water inputs by stemflow. Taniguchi *et al.* (1996) and Germer (2013) also found evidence that stemflow would contribute to groundwater recharge and perched water development in pine and babassu palm forests, respectively. Thus, this kind of concentrated rainwater infiltration from stemflow, serving as a point input of rainwater, has great implications for hydrological processes. Macropores are known to be present around root channels, and they concentrate soil water, which may lead to rapid movements of solutes and pollutants through soils (Beven and Germann, 1982; Kulli *et al.*, 2003). Root systems affect the soil water concentration and movement, which might lead to different rainfall–runoff responses for different tree species, even in

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the same soil type (Jost *et al.*, 2012). Johnson and Lehmann (2006) therefore referred to these effects of stemflow and root channels as the double-funnelling effect of a tree. In consideration of the double-funnelling effect, Martinez-Meza and Whitford (1996) and Li *et al.* (2009) conducted dye experiments, which demonstrated that root channels are the preferential pathways for the movement of stemflow water into soil. Schwärzel *et al.* (2012) conducted a dye experiment and performed detailed analyses that demonstrated the double-funnelling effect of beech trees. They also demonstrated differences in the double-funnelling effect between spruce and beech trees and suggested that stemflow and root-induced preferential flow in a beech forest could trigger fast lateral subsurface flow bypassing large parts of the soil matrix, at least during rainstorm events. On the basis of the aforementioned findings, the double-funnelling effect of a tree has great implications for hydrologic responses in the wetting process.

In addition to the studies discussed earlier, we have clarified several general characteristics of stemflow in relation to the wetting process around a tree on a hillslope in our previous studies. Locally concentrated rainwater inputs, attributable to the stemflow on the downslope side of the tree trunk, caused large and rapid increases in the water content and pore water pressure in the downslope region (Liang *et al.*, 2007). The amount of stemflow for tall *Stewartia* (*Stewartia monadelphica*) was 12–132-fold greater on the downslope side of the stem than on the upslope side. This was likely caused by the uneven area between the upslope and downslope sides of the canopy and by asymmetrical stemflow pathways between the upslope and downslope sides of the trunk (Liang *et al.*, 2009a). The stemflow infiltration process is very different from the vertical rainfall infiltration process. Thus, we developed a three-dimensional model considering the effect of stemflow on soil water dynamics around a tree on a forested hillslope (Liang *et al.*, 2009b). We also demonstrated that a saturated zone still occurred at the soil–bedrock interface underneath the tree in conditions without stemflow supply, which suggests that trees are key locations for rainfall infiltration and the generation of a saturated zone (Liang *et al.*, 2011). On the basis of the observed and simulated results, we suggest that stemflow would have significant implications for hydrological responses in the wetting process.

Ecohydrologic responses could differ between the wetting and drying processes associated with separate rainfall events. In studies investigating soil water demand and water use by vegetation in an ecohydrologic system, the focus has typically been on the vegetation species or density (Garcia-Estringana *et al.*, 2012; Emanuel *et al.*, 2014), root system (Archer *et al.*, 2012; Robinson *et al.*, 2012) or evapotranspiration (Schume *et al.*, 2004; Krämer and Hölscher, 2010). Krämer and Hölscher (2010) found

that soil water was extracted at a higher rate in species-rich plots at the beginning of the drying process, but only beech-dominated stands were able to increase soil water extraction at later stages of the drying process when atmospheric evaporative demand was higher. This suggests that soil water redistribution would not only differ with vegetation type but would also vary during the drying process. Schwärzel *et al.* (2009) estimated the water fluxes of soil water, evaporation and transpiration and found that the differences in evapotranspiration estimated using the eddy covariance and soil water balance methods were often large, which was due to stemflow-induced wetting, and subsequent drying around the trunk could not be monitored. Germer (2013) found that root distributions were related to soil water redistribution and that the infiltration of stemflow, causing the development of perched water tables, would still occur even after small rainfall events. Although many studies have investigated the ecological or hydrological influences of stemflow on soil water dynamics in the wetting process, the process by which stemflow affects soil water redistribution in the drying process has rarely been discussed. In particular, an understanding of the separate involvement of stemflow and root channels in the drying process is still lacking.

The purpose of this study was to clarify the effects of stemflow on soil water depletion around a tree on a hillslope. We used a 2-year data set of rainfall and soil water content to compare soil water depletion in the drying process before and after the interception of stemflow. In addition, we simulated soil water depletion with the three-dimensional Richards equation, which was used to demonstrate rainwater drainage and evapotranspiration in the drying process. With these investigations, we attempted to address the following questions:

- How much soil water depletion occurs around the tree? Does most soil water depletion occur near to the tree?
- How does stemflow contribute to soil water depletion in the drying process?

METHODS AND MATERIALS

Study area

Field observations were made on a hillslope (Figure 1a) at the Kamigamo Experimental Station of Kyoto University, which is located in southern Kyoto Prefecture, central Japan (35°04' N, 135°46' E). The climate of this area is warm temperate. The mean annual air temperature for the period of 1976–2005 was 14.7 °C, with the highest and lowest monthly averages of 27.4 °C (August) and 2.4 °C (January), respectively. Mean annual precipitation was 1523 mm, distributed year-round, with a peak in summer

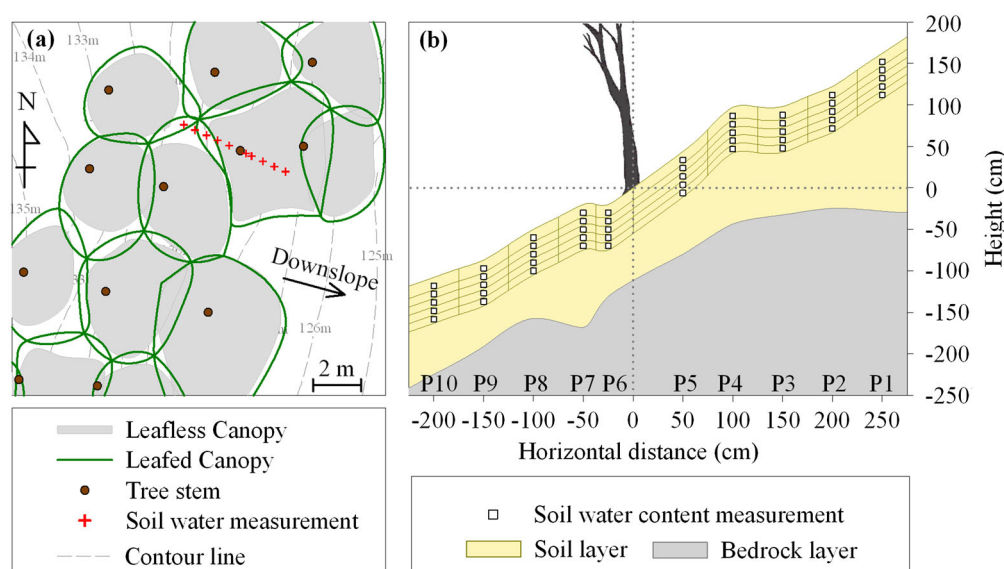


Figure 1. (a) Topography of the study area, showing the locations of tree stems (*Stewartia monadelpha*), leafed and leafless projected canopy areas, and soil water measurement points. (b) Longitudinal section along the observation line for the tree, showing soil water content measurement points P1 through to P10.

and only a few centimetres of snow for a short time in winter.

The hillslope had a mean gradient of 28° , with brown forest soil, classified as Cambisol, underlain by sandstone and slate. There was a small and near-dry channel located 15 m southeast of the hillslope. Tall stewartia (*Stewartia monadelpha*), planted in 1956, was the predominant land cover (Figure 1a). The genus *Stewartia* includes 21 species that are distributed throughout Japan, Korea, China and eastern North America (Li *et al.*, 2002). Of these, tall stewartia is widespread in natural forests in western and southern parts of Japan and is classified by the US Department of Agriculture into hardiness zones 6B through 8B. Like most species of *Stewartia*, tall stewartia is a deciduous broad-leaved tree with upward-tilting branches and a smooth-exfoliated bark. In this region, tall stewartia undergoes leaf fall in November and re-growth begins in April.

Measurements of soil water dynamics

Measurements of soil water dynamics were conducted from 22 April 2007 through to 5 May 2009. To monitor the soil water dynamics around a tree, we selected a tall stewartia tree (Figure 1a; height: 17.47 m; diameter at breast height: 22.3 cm; leafless projected canopy area: 17.39 m^2) and delineated a longitudinal observation line from the upslope to downslope regions of the tree (Figure 1b). No understory vegetation was present along the observation line so that only the effects of the tree on soil water dynamics were observed. No streamflow was observed in the small channel during the observation period, which confirmed

that soil water dynamics were not influenced by saturation from the channel. We installed capacitance meters (EasyAG-5p: Sentek, San Diego, CA, USA) at the following 10 points: 250 cm (P1), 200 cm (P2), 150 cm (P3), 100 cm (P4) and 50 cm (P5) upslope from the tree stem; and 25 cm (P6), 50 cm (P7), 100 cm (P8), 150 cm (P9) and 200 cm (P10) downslope from the tree stem (Figure 1b). Each capacitance meter consisted of five sensors at depths of 10, 20, 30, 40 and 50 cm; therefore, the changes in soil water content at a total of 50 locations was measured. The measurements were simultaneously and automatically recorded at 5-min intervals by a data logger (Campbell, CR1000). The soil depths at all sampling points were estimated by penetration tests to be between 104 and 190 cm (Figure 1b). Gross rainfall data provided by the Kamigamo Experimental Station of Kyoto University was measured using a tipping-bucket rain gauge (RT-5: Ikeda Keiki, Japan; 0.5 mm per tip) recorded at 10-min intervals at an open and flat site, which is located 112 m from the observation slope and has the same elevation as the observation slope.

Data analysis

To determine the effect of stemflow on soil water redistribution, we compared the differences in soil water depletion between the observation period before (SF period, 22 April 2007 through to 8 April 2008) and after (no-SF period, 9 April 2008 through to 5 May 2009) stemflow interception for the tree. In the no-SF period, the intercepted stemflow was measured by a tipping bucket gauge installed at a location 4 m laterally from P10. Thus,

rainwater input attributable to stemflow for the tree was removed from the observation line of soil water dynamics. Stemflow of the other trees that was not intercepted should not affect the observation line because the effect of stemflow on shallow soil layers was limited to the downslope area close to a tree stem (e.g. P6 and P7 for the observation tree) (Liang *et al.*, 2007). Both the SF and no-SF periods were approximately 1 year long. An individual rainfall event was defined as having more than 0.5-mm precipitation and being separated from other events by six consecutive hours without rainfall. To verify the detailed soil water dynamics in the drying process in the SF and no-SF periods, we analysed the spatiotemporal variations in soil water storage (S) at each point and the changes in soil water content (θ) at each location for the duration of each no-rainfall period (Figure 2). The measurements from each of the 50 capacitance meter sensors were assumed to represent θ for each element of the mesh system shown in Figure 1b. For each sampling point, S from the surface through to a depth of 55 cm was calculated by θ measured at depths of 10, 20, 30, 40 and 50 cm. We could then calculate the soil water depletion (WD) at each point for the duration of each no-rainfall period. The WD was defined as S at the start of the no-rainfall period minus S at the end of the no-rainfall period, which was usually positive (Figure 2).

In addition, the simulated WD , calculated using the three-dimensional Richards equation, was compared to the observed WD at each point. In the simulation, different infiltration patterns of throughfall and stemflow in the rainfall redistribution process were disregarded, and

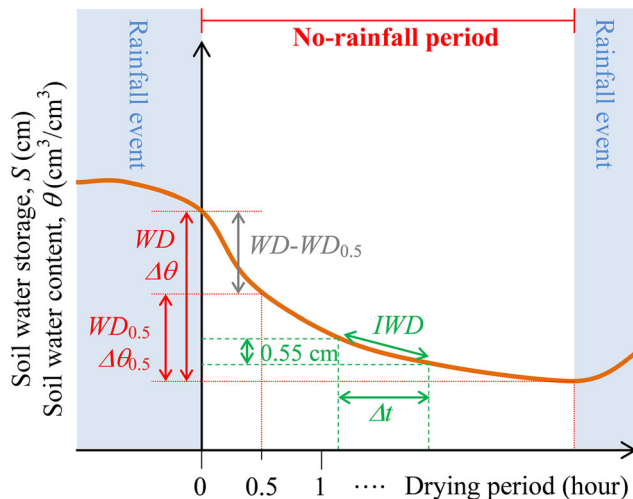


Figure 2. A schematic diagram for the analysis of the soil water redistribution process in the drying period. Soil water storage (S) or soil water content (θ), soil water depletion (WD), soil water content change ($\Delta\theta$), soil water depletion ($WD_{0.5}$) or soil water content change ($\Delta\theta_{0.5}$) from 0.5 h after the start through to the end of each drying duration, and the instantaneous soil water depletion rate (IWD) in the variable time interval (Δt) with soil water depletion greater than 0.55 cm were analysed for each no-rainfall period.

rainfall was evenly inputted across the soil surface area, as the same way as in Model 2 (i.e. the conventional model) described in our previous study (Liang *et al.*, 2009b). After rainwater supplied at an intensity of 45 mm per hour for 1 day, we calculated the WD at each sampling point for 10 days without a rainwater supply. In the simulation procedure, surface evaporation and root uptake were disregarded. Soil hydraulic parameters at each depth and sampling point, except for P6 and P7, were estimated using the data set from the field sprinkling experiment. The dramatic variations in θ were observed at depths of 50 cm at P6 and 30 cm at P7 but were not found at the other sampling points. This was most likely due to the existence of macropores around the tree roots (Liang *et al.*, 2007). The parameters of the saturated soil water content and the residual soil water content at the depths of 50 cm at P6 and 30 cm at P7 were fixed to the maximum and minimum θ values in the observed data sets, respectively. Thus, the effective porosity at P6 and P7 was greater than at the other sampling points (Table I). Other details (e.g. the mesh design, boundary conditions, soil hydraulic parameters estimation procedure and simulation procedure) were described in our previous study (Liang *et al.*, 2009b). On the basis of the aforementioned procedure, the influences of topography and soil hydraulic properties at each sampling point were appropriately reflected in the simulation. Thus, the simulation could approximately represent WD in the drying process, starting from a near-saturated condition in the no-SF period without evapotranspiration.

In addition to determining the soil water depletion WD based on the duration of the no-rainfall period, we also calculated the rate of instantaneous soil water depletion (IWD) during each no-rainfall period in the SF and no-SF periods. The variable time interval Δt was defined as the interval with an average decrease of θ at depths of 0–55 cm of more than 1%, which corresponded to a WD greater than 0.55 cm (Figure 2). Because the IWD would be controlled by S , the IWD was compared with the average S in Δt at each sampling point in the SF and no-SF periods. As a reference, we also calculated the IWD at all sampling points from the simulation described earlier. On the basis of previous investigations and reviews of evapotranspiration in many different types of forest, the

Table I. Average values of effective porosity (θ_e) and saturated hydraulic conductivity (K_s) calculated using the soil hydraulic parameters from the surface through to a depth of 55 cm.

Points	θ_e (cm ³ /cm ³)	K_s (cm/s)
P6	0.865	6.82×10^{-2}
P7	0.835	5.90×10^{-2}
Other points	0.607	3.44×10^{-2}

range of evapotranspiration rates were 0.21–0.30 cm/day (Suzuki, 1985) or 0.11–0.35 cm/day (Komatsu *et al.*, 2008) in Japan and 0.06–0.41 cm/day (Zhang *et al.*, 2001) worldwide. Therefore, we presumed that evapotranspiration rates of 0.1–0.4 cm/day would be a suitable reference range. Through this procedure, we could effectively separate the components of rainwater drainage and evapotranspiration in the *IWD*.

Finally, we investigated occurrences of ‘most soil water depletion near the tree’ (MDNT) for each no-rainfall period by the order of *WD* among all sampling points in the SF and no-SF periods. We assumed that MDNT would be true if the first and second largest *WD*s occurred at the three pairs of P6 and P5, P6 and P7, or P5 and P4 near the tree (Figure 1b). On the basis of the order of *WD* at P1–P10, we could then determine the most representative drying patterns in the SF and no-SF periods.

RESULTS

General trends in soil water dynamic changes

Figure 3 shows the 2-year time series of rainfall events, no-rainfall periods, and soil water content (θ) at all depths and sampling points in the SF and no-SF periods. There were 84 and 116 no-rainfall periods observed within the SF and no-SF periods, respectively.

For the wetting process (i.e. rainfall events), θ increased at all depths in the upslope region when the rainfall intensity was high, with responses at P3 larger than those at the other sampling points in the SF period. When rainfall intensity was low, the responses of θ were confined to the upper layers at depths of 10 to 20 cm. In the downslope region, similar responses of θ to the sampling points in the upslope region were observed for P8–P10, but for P6 and P7, there were dramatic changes in θ even for low rainfall intensity events. In the no-SF period, the response of θ was

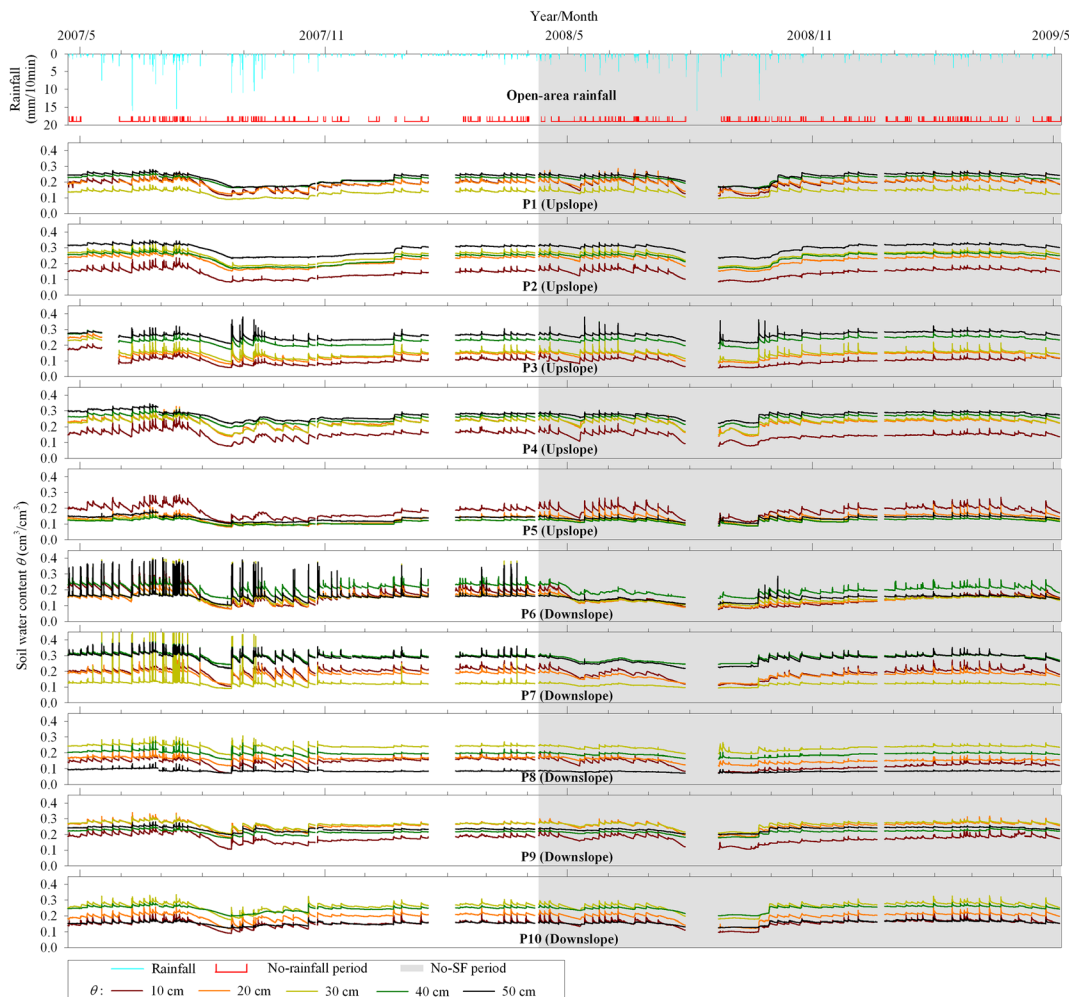


Figure 3. Open-area rainfall, no-rainfall periods and soil water content (θ) at 10-min intervals at ten measurement points (P1–P10) in regions upslope and downslope of the tree. Interruptions within the data lines indicate missing data periods. The shaded period indicates the no-SF period, in which stemflow from the tree was intercepted.

similar at all sampling points in both the upslope and downslope regions, and it varied with rainfall intensity although it was relatively sensitive and large responses of θ at some depths were still found at P3 and P6 when the rainfall intensity was high. The dramatic changes in θ at P6 and P7 disappeared in the no-SF period. This was attributed to the large local water supply from stemflow concentrated on the downslope side of the trunk (Liang *et al.*, 2011).

For the drying process (i.e. no-rainfall periods), θ decreased gradually in winter but more substantially in summer as a result of the active evapotranspiration. In the SF period, θ values decreased immediately at the end of the rainfall events and then decreased gradually during the no-rainfall periods at all sampling points. Particularly dramatic decreases at the end of the rainfall events were observed at P6 and P7. In the no-SF period, soil water depletion patterns were similar at all sampling points. The dramatic decreases of θ at P6 and P7 in the SF period but not in the no-SF period indicates that the large water supply due to stemflow was not retained in the soil layers investigated here but immediately infiltrated to deeper layers in the SF period.

Soil water depletion during no-rainfall periods

Figure 4 shows the soil water depletion WD at 0–55-cm depths during each no-rainfall period in the SF and no-SF periods. In the no-SF period, there were similar variations in WD at all sampling points, with the maximum WD ranging from 1.03 to 2.57 cm. However, in the SF period, WD at P6 and P7 was much greater and varied more dramatically than at the other sampling points. The maximum WD was greater at P6 (5.18 cm), P7 (3.60 cm) and P3 (3.57 cm) than at the other sampling points, with a range of 1.57–2.10 cm, which was similar to that in the no-SF period. We used a logarithmic regression to evaluate the WD at each point in the SF and no-SF periods. Although the R^2 values of the logarithmic regression were low, with a range of 0.04–0.37, the logarithmic regression could show an approximate trend between the WD and no-rainfall periods. On the basis of the regression values of the WD at the tenth day in the no-rainfall period, the WD at all points ranged from 0.44 to 1.59 cm in the SF period and 0.38 to 0.97 cm in the no-SF period. The differences in WD in the regression between the SF and no-SF periods were 0.88 and 0.73 cm at P6 and P7, respectively, and ranged from –0.25 to 0.30 cm at the other sampling points. Thus, the significant differences between the SF and no-SF periods at P6 and P7, with the maximum WD and greatest regression value of WD , were related to the contribution from stemflow.

Figure 4 also shows the simulated WD calculated by the Richards equation, in which the simulated ΔS were divided into eight cases with different initial conditions of 0, 0.5, 1,

3, 6, 12, 24 and 48 h after 1 day of rainwater was applied. This represents soil water depletion starting from different initial conditions from near-saturation to moderately wet in the no-SF period. After drying for 10 days, the simulated WD was 5.34 and 4.49 cm at P6 and P7, respectively, and ranged from 3.42 to 3.83 cm at the other sampling points. A potentially higher WD could be produced at P6 and P7 than at the other sampling points, as a result of the higher effective porosity and saturated hydraulic conductivity at P6 and P7 (Table I). The observed WD at P6 and P7 in the SF period were matched to the simulated values for the initial conditions with high levels of soil moisture (i.e. 0.5 h or later after near-saturation conditions). In contrast, the observed WD at P6 and P7 in the no-SF period were matched to the simulated values for the initial conditions with lower soil moisture (i.e. 3 h or later after near-saturation conditions). It suggests that stemflow would affect the initial conditions at P6 and P7 in the drying process, which would not necessarily be represented if the input of stemflow were disregarded.

On the basis of the average saturated hydraulic conductivity used in the simulation (Table I), the saturated flow infiltrated from the surface to the depth of 55 cm would take 13, 15 and 26 min at P6, P7 and the other sampling points, respectively. Thus, we calculated the observed soil water depletion from 0.5 h after the start through to the end of the non-rainfall period ($WD_{0.5}$), as shown in Figure 2, and presumed that $WD - WD_{0.5}$ could approximately represent the contribution of saturated infiltration flow induced by stemflow (Figure 5). The $WD - WD_{0.5}$ was close to zero for the periods with almost no-rainfall in both the SF and no-SF periods at the sampling points other than P6 and P7. At P6 and P7, $WD - WD_{0.5}$ was close to zero in the no-SF period but varied considerably for many of the no-rainfall durations in the SF period. The $WD - WD_{0.5}$ at P6 and P7 had maximum values of 4.67 and 1.30 cm, respectively, in the SF period but were 0.26 and 0.39 cm, respectively, in the no-SF period. Other than these maximum values, $WD - WD_{0.5}$ at P6 and P7 for many no-rainfall periods corresponded approximately to the differences in WD in the regression between the SF and no-SF periods (i.e. 0.88 and 0.73 cm at P6 and P7, respectively) in Figure 4. This clearly indicated that the effect of stemflow on WD in the downslope area near the tree was limited to the duration of the initial drying period of 0.5 h.

Instantaneous soil water depletion rate during no-rainfall periods

To clarify the detailed variations in WD during the drying process, we analysed the relationship between the average S and IWD during time intervals (Δt) in which WD was more than 0.55 cm during each no-rainfall period within

the SF and no-SF periods (Figure 6). Generally, soil water depletion in the drying process can be attributed to the two hydrologic processes of rainwater drainage (i.e. vertical infiltration) and evapotranspiration (i.e. surface evaporation and root uptake), which are dependent on hydraulic conductivity and weather conditions, respectively. The dependence relationship between the *IWD* and *S* should be significant in the rainwater drainage process, but not in the evapotranspiration process. For rainwater drainage, stemflow or preferential drainage pathways can cause the high *IWD*. Thus, the *IWD* values can be used to evaluate the components in the drying process. On the

basis of the distribution shape and the occurrence of the extremely high *IWD* (greater than 26.4 cm/day; i.e. 0.55 cm per 0.5 h), the sampling points in Figure 6 could be classified into two groups, one with an atypical pattern at P3, P6, P7 and P8, and the second with the typical pattern at the other sampling points.

At the sampling points displaying the typical pattern, the maximum *IWD* was almost lower than 10 cm/day in both the SF and no-SF periods. The *IWD* decreased as *S* decreased in the zone with higher *S* values, but there was little variation in the zone with lower *S*, showing an inflection point between higher and lower *S* at each point.

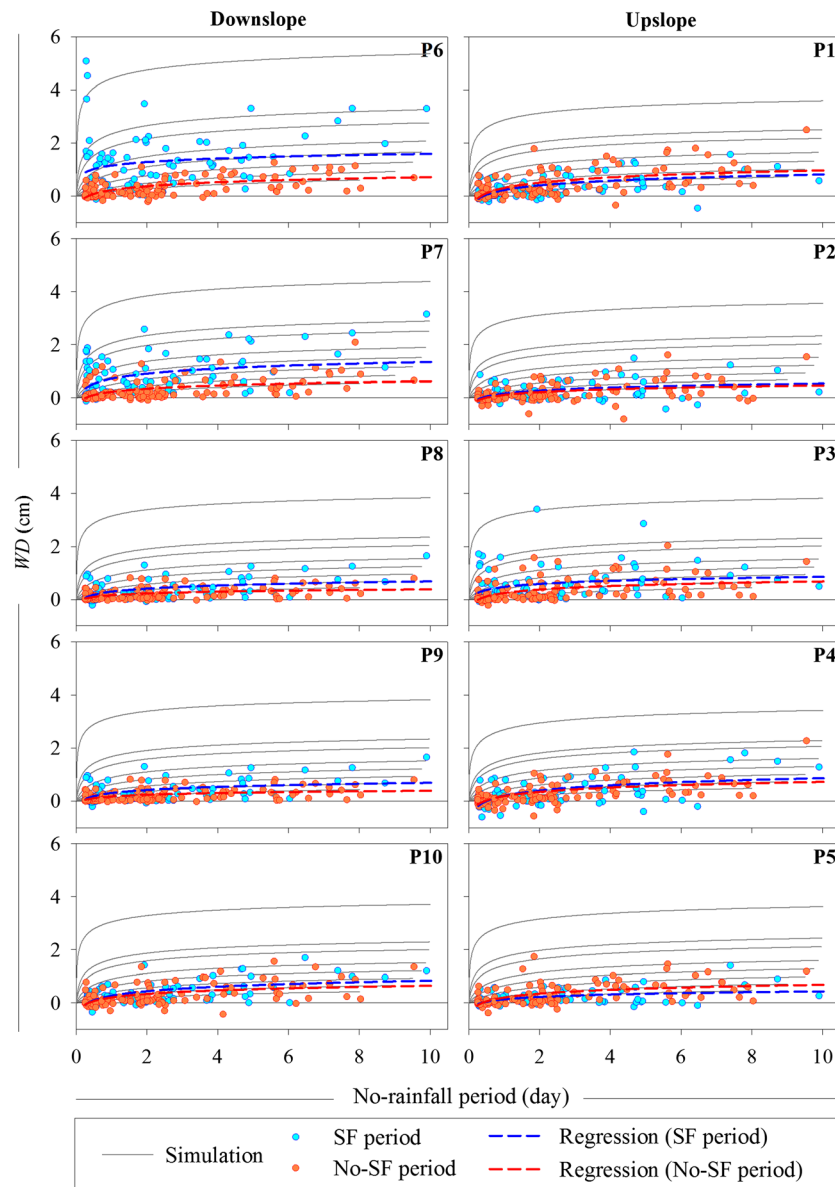


Figure 4. The observed soil water depletion (*WD*, circle) at depths of 0–55 cm for each no-rainfall period within the SF and no-SF periods. Dashed lines indicate logarithmic regressions of *WD* in the SF and no-SF periods. The temporal variations in the simulated *WD* (line) were calculated using different initial conditions of 0, 0.5, 1, 3, 6, 12, 24 and 48 h after the 1 day of rainwater was applied at an intensity of 45 mm per hour (from the top line to the bottom line).

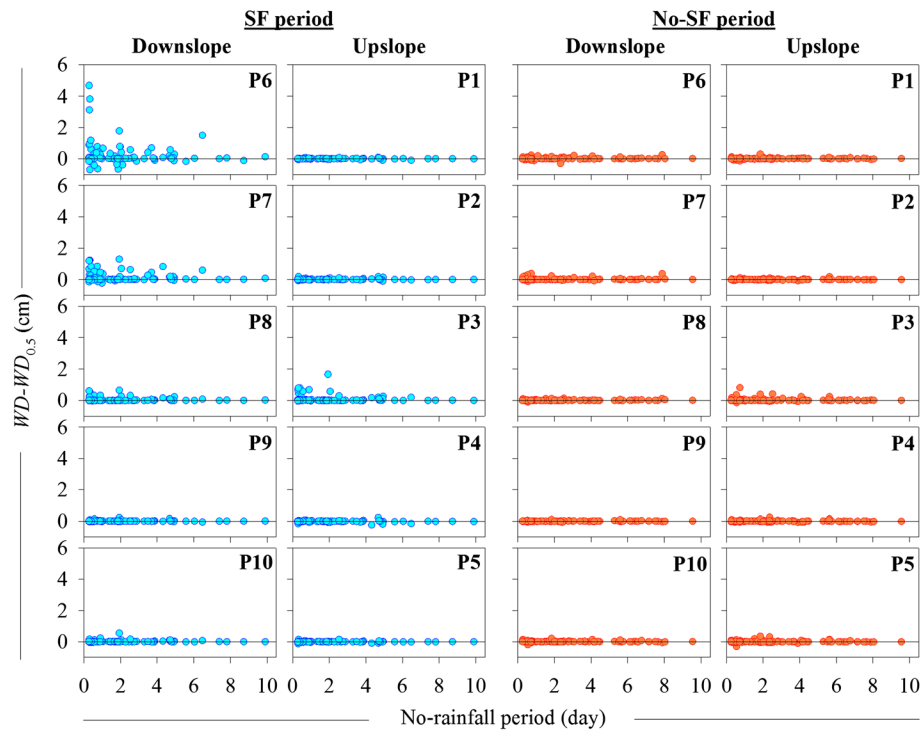


Figure 5. Differences between soil water depletion at depths of 0–55 cm from 0 to 0.5 h after the start through to the end of each no-rainfall period ($WD - WD_{0.5}$) in the SF and no-SF periods.

The inflection point was not found at a consistent S value but located at the same level of the IWD values across the sampling points. On the basis of the reference lines of the simulated IWD at all sampling points, derived from the simulations in Figure 4, the slopes in the relationships of S and the simulated IWD were very close to the observations even though the absolute values did not match. Thus, the observed IWD that decreased as S decreased was mainly attributed to rainwater draining out from the depths at which monitoring occurred. The slight change of the IWD in the zone with lower S was mainly attributed to evapotranspiration and could be matched to the reference lines of the evapotranspiration rates (0.1–0.4 cm/day) estimated in previous studies (Suzuki, 1985; Zhang *et al.*, 2001; Komatsu *et al.*, 2008). Thus, the contribution of rainwater drainage was much greater than that of evapotranspiration in the no-rainfall periods.

For the atypical pattern at P3, P6, P7 and P8, the maximum IWD s were greater than 10 cm/day in the SF period but were almost less than 10 cm/day in the no-SF periods. An extremely high IWD was found at P6 and P7 for many time intervals and at P3 and P8 for some time intervals (i.e. the deep blue circles in Figure 6). Rainwater drainage and evapotranspiration could be effectively separated by the IWD at different S levels for P7 and P8, but not for P6 and P3, where the IWD was more variable. The very large IWD in the zone with higher S contained a contribution from stemflow that increased S

as well as the IWD at P6 and P7. Some of the large IWD values at P8 were likely attributed to stemflow. The variable IWD at P6 and P3 was likely due to preferential drainage pathways that increased the IWD even in the zone with a lower S . This suggests that the atypical pattern could be attributed to stemflow and preferential drainage pathways and that the double effect of stemflow and preferential drainage pathways on the drying process was significant at P6.

Rainwater drainage and evapotranspiration could be easily separated by the dependence relationship between the IWD and S . It should be noted that some sampling points showing the variable IWD values in the zone with lower S (e.g. P5, P8 and P10) could not be exactly classified as the typical or atypical patterns. Nevertheless, both typical and atypical patterns suggest that the contribution of rainwater drainage was much greater than that of evapotranspiration in the no-rainfall periods.

Spatial patterns of soil water depletion

To evaluate the spatial patterns of soil water depletion, we investigated the occurrences of MDNT at each sampling point for each no-rainfall period within the SF and no-SF periods (Figure 7). The order of WD at P1–P10 was classified into three ranks of the most, medium and least depletion (i.e. the red, orange and yellow columns in Figure 7), in which the first and second largest WD s were

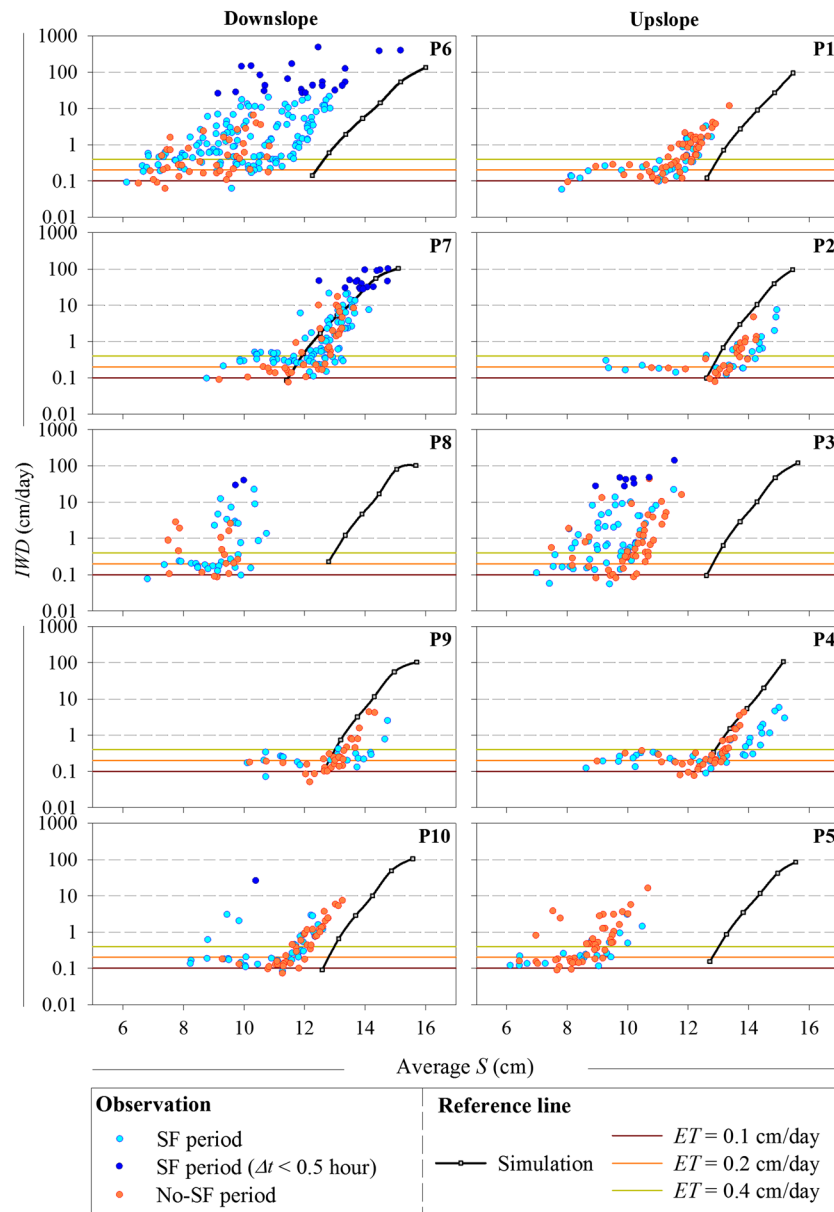


Figure 6. The relationship between average soil water storage (S) and the instantaneous soil water depletion rate (IWD) in the variable time interval (Δt), with soil water depletion greater than 0.55 cm during each no-rainfall period in the SF and no-SF periods. The reference lines indicate the variations in the simulated IWD and evapotranspiration rates (ET) of 0.1–0.4 cm/day. The deep blue circles showing $\Delta t < 0.5 \text{ h}$ indicate the extremely high IWD greater than 0.55 cm per 0.5 h.

assigned to the rank of most depletion (i.e. the red columns in Figure 7). MDNT would be true if the rank of most depletion was observed near the tree. In the SF period, MDNT was true for 55% of no-rainfall periods (i.e. 47/85), in which MDNT occurred mostly at P6 and P7 in the three potential pairs of P6 and P5, P6 and P7, or P5 and P4. The WD increased in the order of P6, P7, P3, P10, P4, P1, P8, P9, P2 and P5, which could be considered as the representative spatial drying pattern in the SF period. In the no-SF period, MDNT was true for 19% of no-rainfall periods (i.e. 22/116), in which MDNT occurred mostly at P6 and P7 in the three

potential pairs. The WD increased in the order of P1, P7, P10, P6, P3, P4, P5, P9, P8 and P2, which was the representative spatial drying pattern in the no-SF period. Thus, MDNT was true for more than half of the cases in natural conditions with stemflow, but the spatial pattern in the drying process would be more random if the contribution from stemflow was removed.

Figure 8 shows the representative drying patterns for two drying periods within the SF and no-SF periods, in which the spatial variations in changes in soil water content changed from the start of the drying period ($\Delta\theta_i$). At each

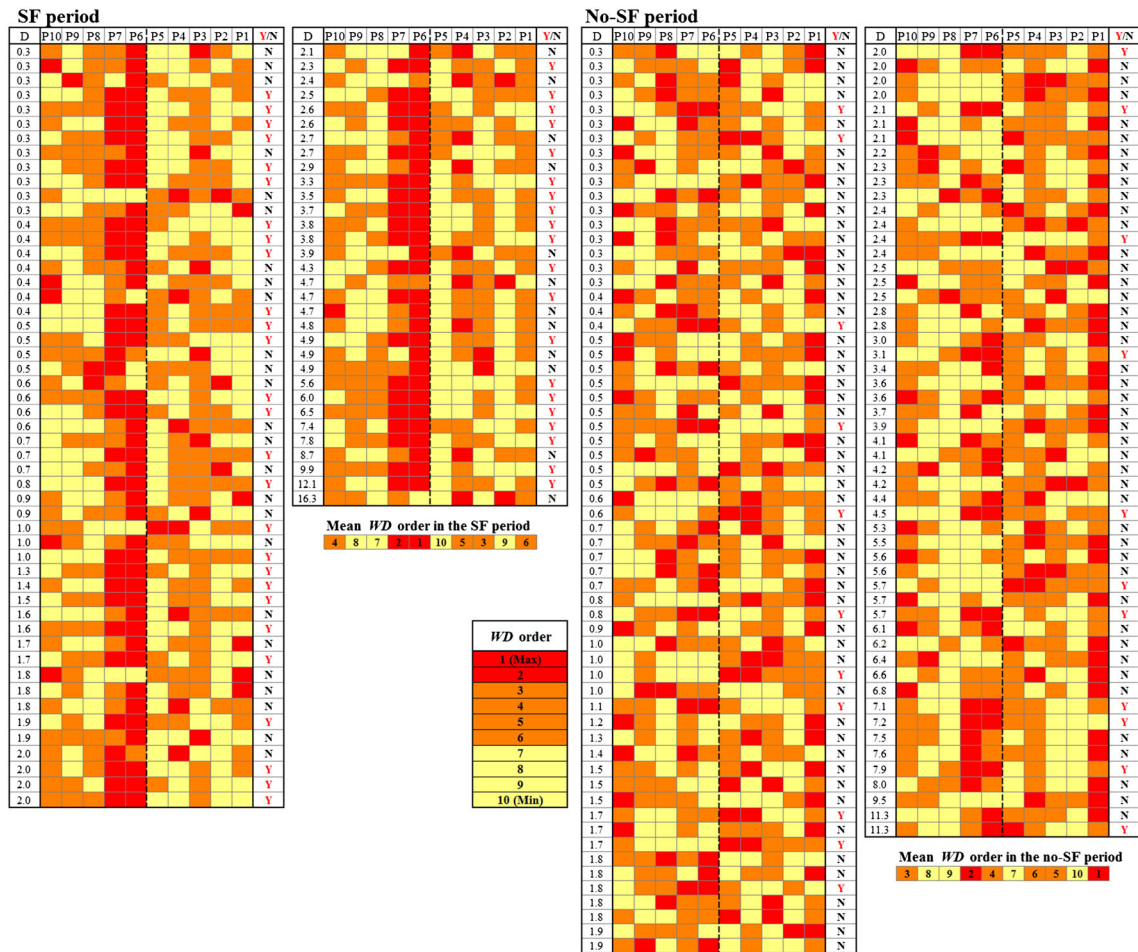


Figure 7. Occurrences of ‘most soil water depletion near the tree’ (MDNT) at points P1–P10 in relation to the drying duration for each no-rainfall period in the SF and no-SF periods. D indicates the number of the drying days for each no-rainfall period. Y or N indicates that MDNT is true or false, respectively. The depletion order indicates the order of soil water depletion (WD) among points P1–P10, in which 1 and 10 indicate the maximum and minimum WD, respectively.

depth data is presented for three time points. For the short drying duration of 34 h from 2007/10/30 to 2007/11/01 in the SF period (Figure 8a), the depletion initially occurred at specific depths at P6 and then expanded to all depths at P6 and P7. However, $\Delta\theta_t$ was small at the other sampling points during the same period. For the long drying duration of 112 h from 2008/03/14 to 2008/03/19 (Figure 8a), the depletion initially occurred at P6 and P7 and then expanded to the other sampling points. Compared with the heterogeneous vertical distributions of $\Delta\theta_t$ at P6 and P7, the other sampling points displayed a relatively even depletion pattern, in which soil water decreased from the surface to deeper layers. Figure 8b shows two long drying periods of 192 h (2008/11/16–2008/11/24) and 197 h (2009/01/23–2009/01/30) in the no-SF period, in which similar spatial and temporal variations in $\Delta\theta$ were measured. Relatively large and rapid $\Delta\theta_t$ were measured in the downslope area rather than the upslope area in the initial and middle stages of these drying periods. Compared

with the initial and middle stages, the depletion expanded to deeper depths at the later stage although the spatial pattern of $\Delta\theta_t$ was not changed. Unlike the other sampling points, heterogeneous vertical distributions of $\Delta\theta_t$ were only found at P6 and P7 where large $\Delta\theta_t$ were measured at specific depths. On the basis of Figure 8a and 8b, we found that large soil water depletion in the downslope area occurred rapidly, in which heterogeneous vertical distributions of $\Delta\theta_t$ were measured at P6 and P7. In contrast, limited soil water depletion occurred relatively slowly in the upslope area. This tendency was more significant in the SF period than in the no-SF period.

Figure 9 shows vertical distributions of the changes in soil water content ($\Delta\theta$) for all no-rainfall periods, indicating where large soil water decreases occurred among the 50 sampling points. The $\Delta\theta$ was defined as θ at the end of the no-rainfall period minus θ at the start of the no-rainfall period, which was usually negative (Figure 2). In both the SF and no-SF periods, the amount of $\Delta\theta$ decreased with

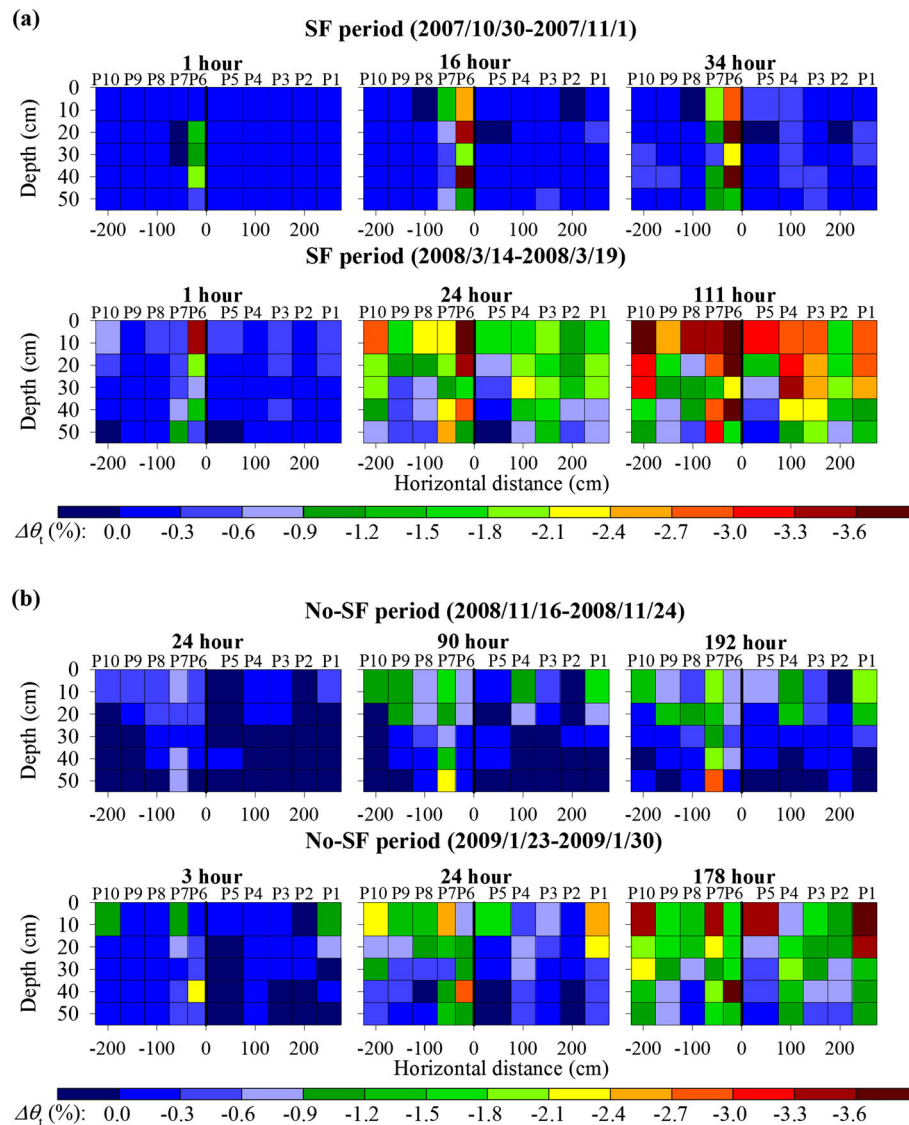


Figure 8. The representative patterns of spatial variations in the changes of soil water content ($\Delta\theta_t$) among the 50 measurement locations for two drying periods within each of the (a) SF period and (b) no-SF period. $\Delta\theta_t$ is defined as the difference between the current water content and the initial water content at the start of drying. The vertical axis of each plot is enlarged to five times the real scale.

depth at all sampling points, except for P6 and P7 where irregular vertical distributions were observed. In the SF period, there was a wide range of $\Delta\theta$ at each depth in P6 and P7, in which there was higher soil water depletion at almost all depths than for the same depths at other sampling points. In the no-SF period, P6 and P7 did not have higher soil water depletion but had averages of $\Delta\theta$ that ranged between -2 and 0% at all depths, with the other sampling points being in the same range. When comparing the averages of $\Delta\theta$ to the changes in soil water from 0.5 h after the start of the no-rainfall period through to the end of the no-rainfall period ($\Delta\theta_{0.5}$, Figure 2), clear differences between $\Delta\theta$ and $\Delta\theta_{0.5}$ were found at only P3, P6 and P7 in the SF period. Additionally, the vertical distributions of $\Delta\theta_{0.5}$ at P6

and P7 in the SF period were more similar to those of $\Delta\theta$ in the no-SF period than those of $\Delta\theta$ in the SF period. Thus, the differences between $\Delta\theta$ and $\Delta\theta_{0.5}$ at P6 and P7 in the SF period were attributed to stemflow. Although the differences between $\Delta\theta$ and $\Delta\theta_{0.5}$ at P3 were only found in the SF period, the irregular vertical range of $\Delta\theta$ (i.e. the box plot in Figure 9) implied preferential pathways at P3, which affected the soil water redistribution process. Moreover, the amount of $\Delta\theta$ and soil porosity derived from undisturbed samples decreased with depth at all sampling points, except for P6 and P7. This lack of $\Delta\theta$ dependence on soil porosity or depth at P6 and P7 in both the SF and no-SF periods was likely due to the influence of preferential pathways (i.e. roots) in the soil water redistribution process.

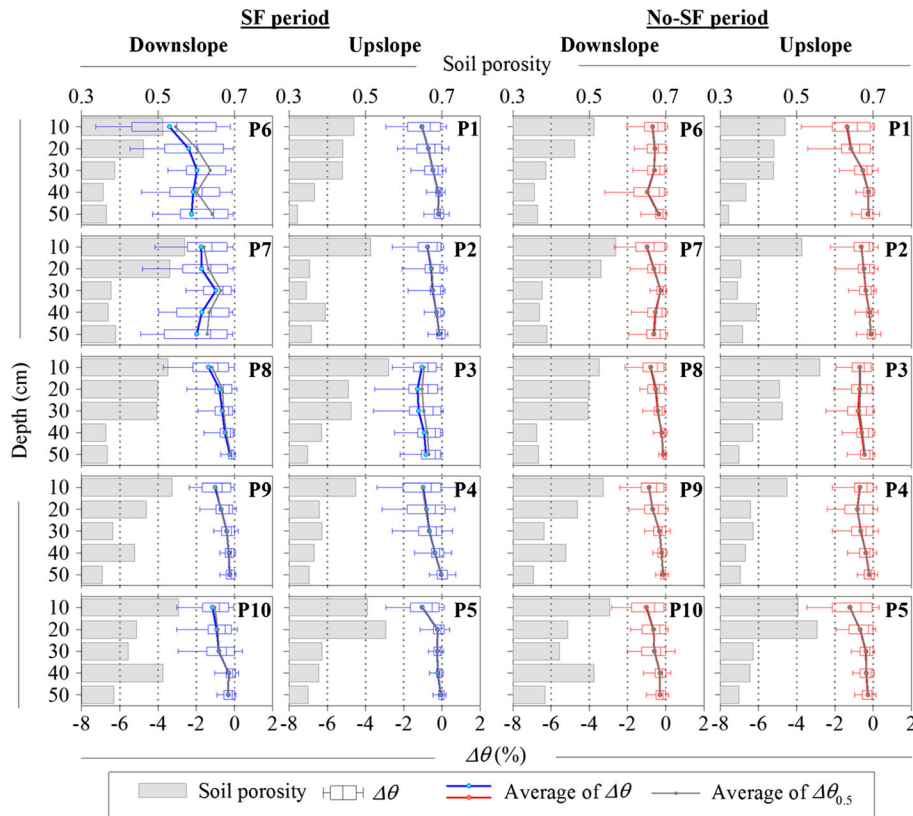


Figure 9. Vertical distributions of soil porosity (bar plot), the variation in water changes ($\Delta\theta$, box plot) and the average of $\Delta\theta$ from 0 ($\Delta\theta$) to 0.5 ($\Delta\theta_{0.5}$) h after the start of the no-rainfall period through to the end of the no-rainfall period within the SF and no-SF periods. Boundaries of the box indicate the 25th and 75th percentiles, the line within the box indicates the median value, and error bars indicate the 10th and 90th percentiles.

DISCUSSION

The double-funnelling effect of a tree on the wetting and drying process

Although many previous studies have investigated or reviewed the effects of stemflow on soil water dynamics (Durocher, 1990; Martinez-Meza and Whitford, 1996; Levia and Frost, 2003; Johnson and Lehmann, 2006), detailed demonstrations based on several rainfall events or no-rainfall periods are still lacking, especially with regard to the separate effects of stemflow and root channels on the soil redistribution process in drying conditions. In our previous work focusing on soil water dynamics in the wetting process (Liang *et al.*, 2011), we observed high soil water storage and irregular variations in the changes of soil water content in the region downslope from a tree in the SF period, which was attributed to the concentration of stemflow on the downslope side of the tree. We conducted sprinkling experiments that separately identified infiltration pathways derived from throughfall and stemflow. The infiltration from throughfall input was limited to shallow soil layers, but stemflow input bypassed surface soil layers and rapidly infiltrated horizontally along thick roots during the same sprinkling period. By focusing on soil water

dynamics in the drying process, we also found high *WD* in the region downslope from the tree (Figure 4). Although substantial soil water depletion was found at specific depths in the region downslope from the tree during the drying process, the dried region would be refilled by large amounts of rainwater from stemflow in the wetting process (Figure 3). This indicates that the regions downslope from a tree were more substantially and iteratively dried and wetted than other regions on a hillslope.

In addition to stemflow, preferential flow also increases the heterogeneity of spatial soil water dynamics. Macropores in the soil layers may be associated with either living or decayed tree roots, which may be very effective at channelling water through the soil layers as preferential flow (Beven and Germann, 1982; Noguchi *et al.*, 1999; Lange *et al.*, 2009). In our previous study (Liang *et al.*, 2011), we found that the frequency of bypass flow occurrence varied little during the wetting process between the SF and no-SF periods. Although dramatic variations in soil water dynamics disappeared in the no-SF period, asymmetrical soil water response patterns were also observed, which were mainly attributed to root-induced bypass flow. In this study, we also found irregular variations in the changes of soil water content at P6 and

P7 during the drying process even in the no-SF period (Figures 8b and 9). This indicates that the soil water redistribution process and spatial distribution of soil water would also be significantly affected by root channels.

The dramatic *WD* in the downslope area near to the tree was attributed to the double-funnelling effect of a tree by stemflow and root-induced pathways (Figures 4 and 6). The effect of stemflow was significant in a short period (0.5 h) after the end of a rainfall event (Figure 5) and raised the initial soil water content at the start of the drying process. The effect of root-induced pathways would enhance the drainage ability during the drying process (Figure 4) and resulted in an irregular vertical distribution of changes in soil water content (Figure 9). Although the root-induced pathways still existed in the no-SF period, a dramatic *WD* in the downslope area near to the tree was not detected. This indicates that a substantial rainwater supply by stemflow was required to produce a high *WD*, even at sampling points with potentially high drainage ability. Thus, we suggest that the double-funnelling effect of a tree is valid not only for the wetting process but also for the drying process and results in dramatic variations in soil water dynamics that are limited to the stemflow-influenced and root-influenced areas of a forested stand.

Although the simulation could represent *WD* at P6 and P7 in the no-SF period when considering root channel influence and adjusting the hydraulic parameters for the area near the tree, it was insufficient for representing *WD* at P6 and P7 in the SF period because the point input of stemflow was disregarded in the simulation (Figure 4). Using the model proposed in our previous study (Liang *et al.*, 2009b), in which stemflow was parameterized as a source flux in the soil layers, would enable a representation of the double-funnelling effect of a tree on soil water dynamics in both the wetting and drying processes.

Effect of stemflow on soil water redistribution processes

In addition to meteorological and geological factors, many previous studies have considered the effects of vegetation on water redistribution processes by different species, evapotranspiration or interrelations of these variables. Ludwig *et al.* (2005) reported that considerably more soil water was stored in small groves of mulga trees than in the intergrove areas where no trees existed, resulting in a deeper wetting front under the mulga grove. Robinson *et al.* (2012) investigated the spatiotemporal variability in hydraulic redistribution within an oak–pine forest and found that the soil moisture demand and usage was greater in the areas with trees than non-tree areas, which indicated that root zones are likely retaining and regulating moisture through hydraulic redistribution or hydraulic lift. In the measurements of Tromp-van Meerveld and McDonnell (2006), transpiration was found

to be limited by soil moisture in areas with a shallow soil depth but in the areas with a deeper soil depth. They also indicated that the spatial differences in soil depth and the total water available were responsible for the observed spatial differences in basal area and species distribution on the hillslope. Schume *et al.* (2004) measured a higher percentage of water in deeper soil layers in a beech stand than in a spruce stand, which resulted from root systems producing different evapotranspiration rates between the beech and spruce stands. Schwärzel *et al.* (2009) demonstrated that the spatial and temporal variability in soil water depletion and transpiration in beech and spruce stands was mainly due to the differences in soil properties and root intensity. The different root architecture and depth of root development between beech and spruce stands not only influenced transpiration but also affected soil water redistribution by preferential flow (Schwärzel *et al.*, 2012). Jackson *et al.* (2000) reported that the soil water content close to the trees increased to a greater extent than at a distance of 2.5 m, even after rainfall, which was attributed to stemflow. They assumed that the higher root activity close to the trees would increase soil water uptake at these locations. On the basis of the conclusions of these studies, it is clear that soil water usage by vegetation is strongly related to the root system, which then affects evapotranspiration in forested stands.

Although many previous studies have indicated the effects of root uptake or evapotranspiration on water redistribution, the contribution of rainwater drainage on soil water depletion was demonstrated to be greater in this study. Soil water depletion in the drying process was mainly attributed to rainwater drainage and evapotranspiration. Generally, soil water depletion at locations near to a tree is expected to be greater than in other areas because of obvious root uptake (i.e. transpiration) in the drying process. However, in our present study, MDNT was only true in the SF period (Figure 7). On the basis of calculations of the *IWD* in the no-rainfall periods, we were able to separate rainwater drainage and evapotranspiration and found that the contribution of rainwater drainage was much greater than evapotranspiration at all points in short term (i.e. up to 10 days) no-rainfall periods (Figure 6). The substantial soil water depletion near to the tree was mainly attributed to drainage of the saturated flow from stemflow, rather than evapotranspiration at depths of 0–55 cm. The *IWD* at sampling points near to the tree were no higher than at the other sampling points in the no-SF period, suggesting that the effects of root uptake or transpiration on soil water depletion were not significant when drainage was still occurring in no-rainfall periods. Thus, we suggest that stemflow raises the initial soil moisture at the start of a drying period and therefore causes substantial soil water depletion near to the tree. It should be noted that the contribution of rainwater

drainage to soil water depletion is greater than that of evapotranspiration only in the short duration no-rainfall periods between rainfall events. Because the amount of *WD* was dependent on the initial conditions and stemflow could have an effect on *WD* within a short time in the initial 0.5 h of the SF period (Figure 5), the analysis based on the no-rainfall periods in Figure 4 does not necessarily represent the *IWD* in Figure 6. Likewise, the dramatic and instantaneous variations in *WD* would be weakened or averaged by analysis units of a longer period, such as monthly or annual terms.

CONCLUSIONS

We used a 2-year data set of soil water content around a tree on a hillslope to analyse soil water depletion in the drying process, before (SF period) and after (no-SF period) intercepting stemflow. We attempted to address how soil water depletion occurred around the tree, where most soil water depletion occurred around the tree, and how stemflow contributed to soil water depletion in the drying process. The results are summarized as follows.

1. Dramatic soil water depletion was observed at sampling points in the downslope area near to the tree and was attributed to the double-funnelling effect of a tree by stemflow and root-induced pathways. The effect of stemflow was significant for a short time (0.5 h) after the end of rainfall events, which increased the initial soil water content at the start of the drying process. The effect of root-induced pathways was to enhance drainage during the drying process, which resulted in irregular vertical distributions of soil water.
2. Although substantial soil water depletion was found at specific depths in the region downslope from the tree during the drying process, the dried region was refilled by large amounts of rainwater due to stemflow in the wetting process. We suggest that the double-funnelling effect of a tree is valid not only for the wetting process but also for the drying process, resulting in dramatic variations in soil water dynamics that are limited to the stemflow-influenced and root-influenced areas of a forested stand.
3. Soil water depletion at locations near to a tree would be expected to be greater than in other areas due to root uptake in the drying process. This was true for more than half of the cases in the natural condition with stemflow, but the spatial pattern in the drying process was more random if the contribution from stemflow was removed. This was due to the much larger instantaneous soil water depletion rates due to rainwater drainage rather than evapotranspiration in short duration no-rainfall periods between rainfall events.

Unlike many previous studies that have indicated the effect of root systems on transpiration and therefore soil water usage by vegetation, this study demonstrated the substantial effect of stemflow on soil water depletion in short duration no-rainfall periods. Thus, our findings enable an understanding of the separate involvement of stemflow and root channels in the drying process, under a warm temperate climate.

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